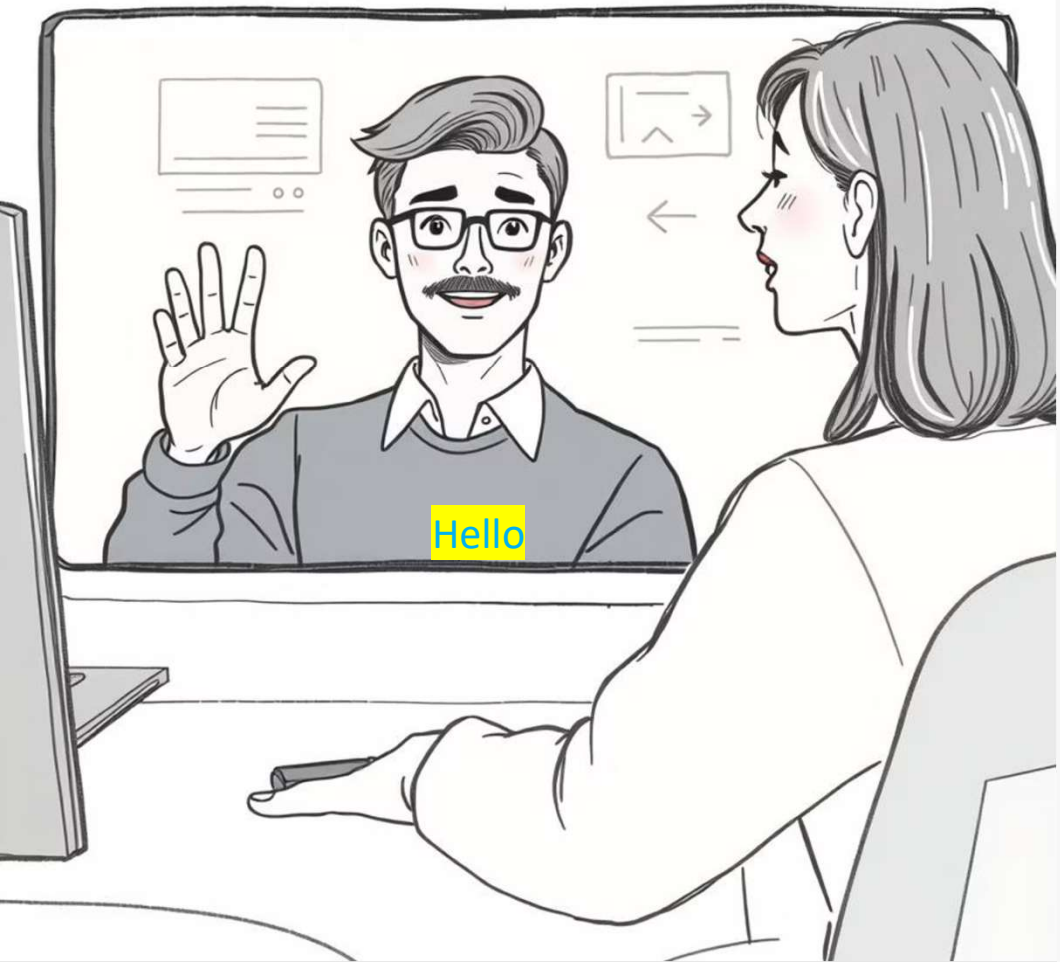

Sign to Speech, text Video Call Platform

Students

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Guide

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Problem Statement

Effective communication between the deaf and hearing communities is frequently hindered by technological fragmentation and the lack of a comprehensive, real-time solution. Existing assistive technologies typically operate as isolated applications, offering tools for either real-time speech-to-text transcription or offline sign language recognition. This isolated approach prevents spontaneous and seamless conversations, forcing users to rely on a collection of disparate tools or a human interpreter to bridge the communication gap.

Specifically, there is a critical absence of a unified, multi-modal platform that integrates these functionalities within a single video conferencing environment. The core problem this project addresses is the lack of a real-time system capable of translating speech to sign language, and vice-versa, within a single video stream, thereby failing to support bi-directional, natural dialogue. The absence of such a cohesive system limits social inclusion and professional opportunities for the deaf community and creates a significant barrier to accessible communication.

Literature Survey

S.N	Title of Paper	Description	Merits	Demerits
1	**Video Chat using WebRTC with In-built Sign Language Detection**	Uses WebRTC and TensorFlow for real-time sign detection in video calls to support speech/hearing-impaired users.	Real-time detection, integrated with video chat.	Detection accuracy drops in low lighting or poor bandwidth.
2	**Sign Language Based Video Calling App**	Converts static ASL gestures into text using a clear pipeline of ML steps.	Well-structured method, clear pipeline.	Limited to static gestures; lacks dynamic sign support.
3	**A Deep Sign Language Recognition System for Indian Sign Language**	Deep learning system for recognizing ISL gestures with improved accuracy.	High accuracy with deep learning.	Needs large dataset; weak at sentence-level recognition.
4	**Sign Language Multimedia-Based Interaction for Aurally Handicapped People**	Web-based system offering sign videos, captions, and audio.	Multi-modal design, accessible.	No real-time interaction; static content.
5	**Sign Language Translator System Using CV and LSTM**	Uses LSTM and NLP to convert sign language to text in real-time.	Real-time; NLP-enhanced; inclusive.	Dataset limitations, sensitive to environment.
6	**Real Time Sign Language Translator Using ML**	Translates sign to speech and vice versa using MediaPipe and ML in a web app .	Multi-language support, includes learning modules.	Accuracy varies with dataset and accents.
7	**Real-Time ASL Interpretation Using Deep Learning and Keypoint Tracking**	Uses MediaPipe and DL for accurate, low-latency real-time ASL translation.	High accuracy; works in real time.	Performance affected in varied environments.
8	**Real-time Conversion of Sign Language to Text and Speech, and vice-versa**	Bi-directional system with webcam and animated avatar for sign ↔ speech.	Comprehensive communication tool.	Depends on avatar realism and webcam quality.

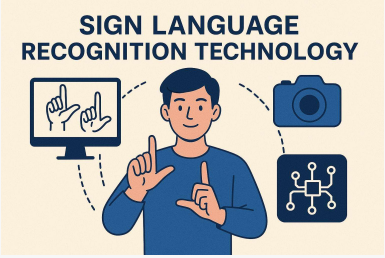
Existing System Overview

1. Video Calling Platforms

Applications like Zoom, Google Meet, and WhatsApp provide real-time video communication
Limitation: No built-in support for sign language recognition or translation

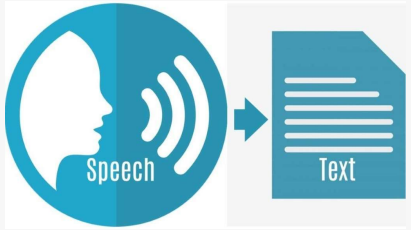
2. Sign Language Learning / Recognition Systems

Systems using AI/ML detect sign gestures and convert them into text or speech
Examples include MediaPipe-based and deep learning-based models
Limitation:
- Mostly limited to **static gestures**
- Lack **real-time integration with video calls**



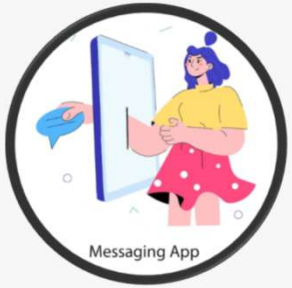
3. Speech-to-Text Systems

Tools like Google Speech-to-Text and Whisper convert speech into text
Limitation:
- Only **one-way communication (speech → text)**
- No conversion into sign language



4. Text-Based Communication Systems

Chat applications enable communication through typing
Limitation:
- Not natural or efficient for real-time conversations
- Not suitable for users dependent on sign language



Existing systems either support communication or translation **but not both together in real time**.

Technologies Used

Frontend
React
WebRTC
React Three Fiber
Socket IO
Tailwind CSS

ML Models
Sign Recognition: MediaPipe & ST-GCN.
Speech-to-Text: Whisper or Google speech API.
CNN & RNN

Deployment
Deployment using Docker on AWS.

Backend
Node.js
MongoDB

PROPOSED SYSTEM & METHODOLOGY

Incremental System Integration:

Combines **sign recognition, speech processing, and real-time video communication** on one platform.

Key Steps:

- **Data Collection** - Capture ISL datasets, extract hand/face/body key points (MediaPipe).
- **Sign Recognition** - Use **ST-GCN / Transformer models** for gesture-to-text conversion.
- **Speech Recognition (ASR)** - Convert speech to text (Whisper / Cloud APIs).
- **Text-to-Speech (TTS)** - Convert recognized text to natural speech (Google/Azure/Coqui).
- **Text-to-Sign** - Map transcripts to pre-recorded sign clips / 3D avatars with expressions.
- **Video Call Integration** - WebRTC for real-time audio, video, captions & sign animations.
- **Backend** - Node.js & Express, MongoDB for storage, ML services via Python/FastAPI & Docker.
- **Testing & Optimization** - Latency <500 ms, model quantization (ONNX/TensorRT), user testing.
- **Deployment** - Frontend (Vercel), Backend (Render), ML (AWS GPU), containerized with Docker.

Outcome: A **scalable, low-latency, real-time bi-directional communication framework** enabling seamless interaction between deaf/hard-of-hearing and hearing users.

Hardware Requirements

Laptop/PC: A system with basic processor (I5), RAM (Min 8 GB)
Camera: HD Webcam (for sign detection)
Microphone: Built-in or external mic (for speech input)
Internet: Stable connection (≥10 Mbps for real-time video for good experience)

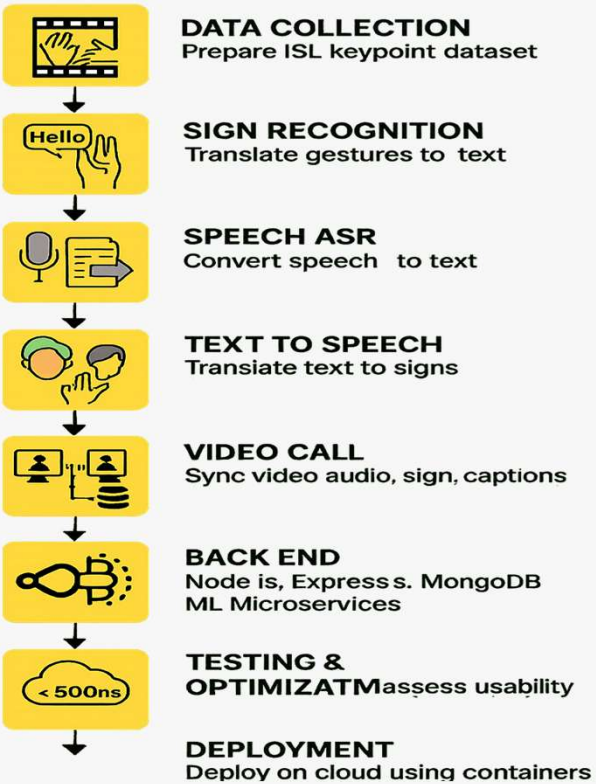
Software Requirements

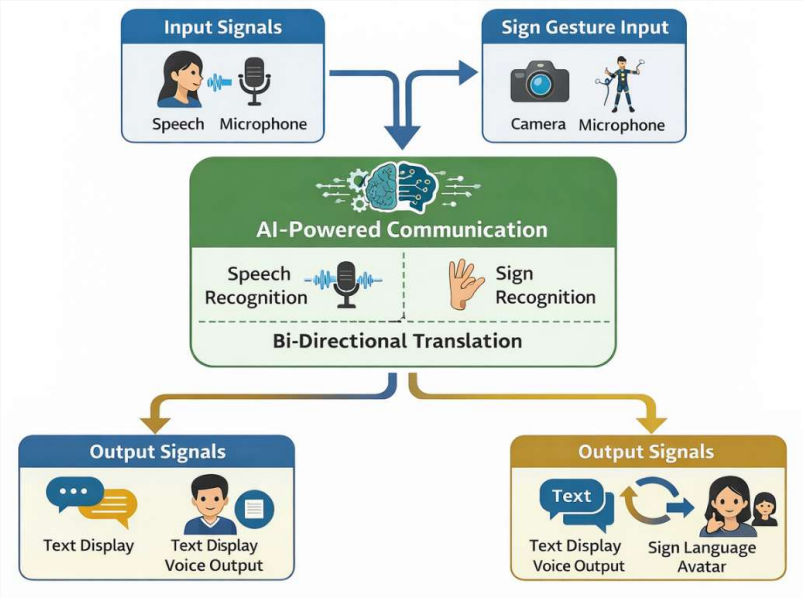
Frontend: React.js, Tailwind CSS Backend: Node.js, Express.js
Database: MongoDB Real-Time Communication: WebRTC, Socket.IO
Machine Learning: MediaPipe, ST-GCN, CNN, LSTM
Speech Processing: Whisper / Google Speech API, Text-to-Speech APIs
3D Rendering: React Three Fiber (for sign avatars)
Deployment Tools: Docker, AWS / Vercel / Render

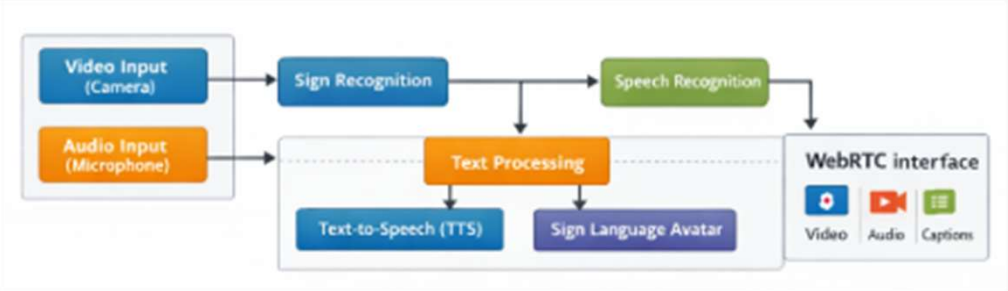
List of Modules

- **User Authentication Module** - Handles user login, signup, and session management
- **Video Calling Module** - Enables real-time video/audio communication using WebRTC and Socket.IO
- **Sign Recognition Module** - Detects sign gestures using MediaPipe and ML models, converts to text
- **Speech Recognition Module** - Converts speech input into text using Whisper / APIs
- **Text-to-Speech Module** - Converts recognized text into audio output
- **Text-to-Sign Module** - Displays sign language using avatars or pre-recorded animations
- **Backend & Database Module** - Manages APIs, real-time communication, and data storage (MongoDB)

Diagrams of flow & others







Results and Discussions

Results

- Achieved **92.5% accuracy** using ST-GCN model for sign recognition
- Real-time performance achieved with:
  - Sign to Speech latency: ~480 ms
  - Speech to Sign latency: ~620 ms
- System supports **bidirectional communication** in live video calls

Discussion

- ST-GCN performs best because it captures **spatial + temporal features of gestures**
- System successfully integrates:
  - Sign recognition
  - Speech processing
  - Real-time video communication
- Compared to existing systems:
  - ✓ Supports **two-way communication**
  - ✓ Works in **real-time video environment**
  - ✓ Combines multiple technologies into one platform

Conclusion

Based on the reviewed papers, most systems focus either on **sign language recognition** or **speech translation**, but not both in **real time**. Some prioritize **accuracy using deep learning**, while others focus on **multimedia content** or **education-based tools**.

However, **very few** offer a **complete, real-time, two-way communication system** suitable for live video calls.
Our project is unique because it:

- Combines **sign recognition**, **speech-to-text**, and **text-to-sign** using avatars
- Integrates all functions into a **single real-time video call system**
- Offers **bidirectional translation**, not just one-way conversion
- Solves key issues like fragmented tools, platform limitations, and real-time responsiveness
- This makes it an **inclusive, practical, and scalable solution** that bridges the gap between the hearing and deaf communities in a way that no single system in the literature currently does.

Future Enhancements

- Support for multiple sign languages and regional variations
- Improved accuracy using advanced models like Transformers and BiLSTM

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